

Coalescence and Translation: A Language Model for Population Genetics

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Probabilistic models such as the sequentially Markovian coalescent (SMC) have long provided a powerful framework for population genetic inference, enabling reconstruction of demographic history and ancestral relationships from genomic data. However, these methods are inherently specialized, relying on predefined assumptions and/or limited scalability. Recent advances in simulation and deep learning provide an alternative approach: learning directly to generalize from synthetic genetic data to infer specific hidden evolutionary processes. Here we reframe the inference of coalescence times as a problem of translation between two biological languages: the sparse, observable patterns of mutation along the genome and the unobservable ancestral recombination graph (ARG) that gave rise to them. Inspired by large language models, we develop *cxt*, a decoder-only transformer that autoregressively predicts coalescent events conditioned on local mutational context. We show that *cxt* performs competitively with state-of-the-art MCMC-based likelihood models across a broad range of demographic scenarios, matching their accuracy in-distribution and approaching it in out-of-distribution settings, with the potential for further improvement via targeted fine-tuning. Trained on simulations spanning the *stdpopsim* catalog (1–3), the model generalizes robustly and enables efficient inference at scale, producing over a million coalescence predictions in minutes. In addition *cxt* produces a well calibrated approximate posterior distribution of its predictions, enabling principled uncertainty quantification. We apply *cxt* to population genomic data from both humans and mosquitoes, highlighting the model’s ability to deal with the complexities of empirical data.

Language models | coalescent theory | uncertainty | *stdpopsim* | simulation-based inference

The stochastic process of genetic lineages tracing back to common ancestors over time gives rise to a complex structure known as the Ancestral Recombination Graph (ARG), with branches that merge through coalescence or divide through recombination (4–7). In the absence of mutations, these branches remain entirely concealed. However, mutations along the branches of the genealogy offer only a noisy, indirect window into this hidden ARG, carrying partial information about topology and the timing of events. Currently the field of population genetics is undergoing a revolution in our ability to infer the ARG that underlies a sample, and over the past few years we have obtained our first glimpses into reconstructions of whole genome genealogies (8–11).

Inferring the ARG (or approximations of it) has long been a central goal in population genetics. One of the earliest attempts was by Griffiths and Marjoram (12), who used Markov chain simulations of the coalescent process for ancestral inference. Subsequent work conceptualizing recombination as a point process along the genome (13), shifted focus toward emphasizing the spatial aspects of recombination, and laid the groundwork for more efficient approximations of the coalescent with recombination such as the sequentially Markovian coalescent (SMC) (14). The SMC approximation has been pivotal in the development of numerous inference methods, beginning with approaches requiring only a single diploid genome to investigate demographic changes, such as PSMC (15) (which effectively infers an ARG for two haploid samples).

Significance

cxt is a language model for population genetics which introduces next-coalescence prediction as translation from observed mutations to coalescence times by modeling the coalescent with recombination as a conditional stochastic process. It learns implicit priors from *stdpopsim* and generalizes across both known and novel demographies. *cxt* generates millions of TMRCA estimates in minutes and samples well-calibrated posteriors for uncertainty quantification. A simple post-hoc correction aligns predicted diversity with the species mutation rate, ensuring robustness to novel evolutionary scenarios.

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Since then, SMC-type methods have been extended in numerous ways including: to multiple genomes (16, 17), integration of site-frequency spectrum information (18), the inclusion of species-specific life-history traits (19–21), as well as algorithms that more efficiently scale as a function of sample size (22–24).

While the SMC approximation enables tractable likelihood-based inference for a restricted class of models, it becomes increasingly difficult to extend when key evolutionary processes, such as population structure, selection, or complex demographic events are introduced. Simulation-based inference (SBI) offers a flexible alternative by bypassing explicit likelihoods and instead learning mappings from data to model parameters directly from simulated datasets. Approximate Bayesian computation (ABC) is a widely used SBI framework in population genetics (25), and has been successfully applied to a range of inference problems, including demographic history reconstruction (26, 27).

However, classical ABC methods rely on reducing data to low-dimensional summary statistics in order to mitigate the curse of dimensionality, often discarding information relevant to inference in the process. This limitation is particularly acute for sequence-based problems involving recombination, where genealogical information is distributed across long, high-dimensional haplotypic sequences. In such settings, exemplified by PSMC-like models, information loss during data compression can substantially limit predictive accuracy and robustness, motivating the development of inference approaches that operate directly on high-dimensional sequence data (see, however, (28)).

A powerful alternative to standard ABC is to utilize deep learning (DL) for simulation-based inference, which has proven broadly effective across diverse domains (29). In population genetics, DL-based methods have been successfully applied to a wide range of model-based inference tasks (30), leveraging standard neural network architectures including fully connected networks (31, 32), convolutional neural networks (33–40), recurrent neural networks (41), and variational autoencoders (42). However, most existing DL approaches in population genetics are trained for narrow, task-specific objectives, such as estimating individual demographic parameters, and often fail to generalize beyond the distribution of their training simulations. While domain adaptation strategies can partially mitigate this limitation (43), they do not fundamentally address the challenge of learning transferable representations of the underlying evolutionary process.

More recently, transformer-based language models have been introduced that enhance prediction capabilities through the learning of stochastic processes, rather than scenario-constrained parameter estimation (40). Inspired by pretrained generative models such as GPT (44), we leverage this paradigm with a small model (10–20M parameters) that translates mutational patterns across chromosomes to coalescent time estimates, opening the door to a generalizable approach: pretraining on diverse coalescent simulations, followed by fine-tuning for specific evolutionary tasks if necessary.

In this work, we introduce *cxt*, the first language model for coalescent estimation to our knowledge. *cxt* is a decoder-only transformer model inspired by GPT-2 and

designed for pairwise time to the most recent common ancestor (TMRCA) prediction—a core component of genealogical inference. In contrast to DNA language models such as GPN-MSA or Evo 2 (45, 46) that are pretrained on the task of masked base prediction from a collection of sequenced genomes, our pretext task leverages synthetic data simulated under explicit population genetic models to predict the *next* pairwise TMRCA along a sequence. Rather than modeling DNA sequences itself, *cxt* aims to recover latent genealogical structure, providing a flexible and generalizable model for downstream inference. To enable this inference, we define a novel task analogous to *next-token prediction*, which we term *next-coalescence prediction*: the model autoregressively predicts the next pairwise coalescence time conditioned on local mutation densities within a fixed context window. While this does not reconstruct full genealogical topologies, it provides a structured translation from observed mutation patterns to pairwise coalescence times. Conceptually, *cxt* is a close neural analog of SMC++: prediction is centered on a pivot (distinguished) pair, with information shared across haplotypes through the SFS.

To evaluate this approach, we focused on three criteria: (1) **accuracy**, performance comparable to or better than recent theory-driven methods such as *Singer* and SMC++; (2) **robustness**, particularly under model misspecification and out-of-distribution scenarios; and (3) **capacity**, the ability to generalize across a diverse range of simulated evolutionary scenarios.

Using extensive simulations and applications to human and mosquito genomic variation data, we show that *cxt* provides competitive performance for inferring local coalescence times across recombining chromosomes, with efficient parallel GPU inference. The generative nature of *cxt* enables rapid sampling of an approximate posterior over TMRCA trajectories, providing uncertainty estimates that are well calibrated. Across a wide range of models, including nearly the full *stdpopsim* catalog, *cxt* maintains reasonable out-of-sample performance, though with some loss in accuracy relative to likelihood-based methods, and can be further improved via targeted fine-tuning. Finally, we apply *cxt* to large-scale human and *Anopheles* population genomic data (47–50), recovering known coalescent-time outliers in humans and time-resolved signals consistent with insecticide resistance dynamics in mosquitoes.

Results

We begin by presenting a conceptual schematic that adapts the general language modeling paradigm to population genetics, and draw parallels with SMC-based coalescent modeling to define our next-coalescence prediction task (Figure 1). We then benchmark *cxt*'s accuracy against current fast inference methods, specifically *Singer* and SMC++ (11, 18). Next, we evaluate the model's cross-scenario generalization capabilities, training and evaluating using the nearly complete *stdpopsim* v0.2 and v0.3 catalogs (1, 2) across diverse species, demographic scenarios, and genetic maps (excluding only bacteria and two species with extreme recombination rates that preclude large-scale simulation and three models with ancient samples). We assess the global accuracy of TMRCA prediction by comparing true and estimated coalescence rates across

simulations. Finally, we apply **cxt** to real genomic data, estimating coalescent times across two chromosomes from the 1000 Genomes Project, using pairs of haploid sequences of British ancestry as our focal samples and chromosome 2L of *Anopheles* from samples taken from Cameroon, Mali, Burkina Faso, Ghana and Uganda.

Fast and Accurate Inference. We developed **cxt**, a decoder-only transformer architecture adapted for the task of inferring local coalescence times from mutation patterns across recombining genomes. Unlike language models that embed discrete tokens, **cxt** projects continuous mutation densities into a latent space using feed-forward networks, followed by attention-based contextualization. Rotary positional embeddings encode physical distance along the genome, allowing the model to capture locality in genealogical relationships. Trained using a next-coalescence prediction objective, **cxt** captures the stochasticity of evolutionary processes while achieving rapid and accurate inference.

We show a descriptive schematic of the model in the left panel of Figure 1. **cxt** follows a general language modeling paradigm in which the model predicts a discretized coalescent time for each genomic window based on preceding predictions and the local mutational context. Specifically, we divide the genome into fixed-size positional windows and discretize coalescent times into bins on a logarithmic scale. Given a sequence of ground truth discretized pairwise TMRCA values $E = (e_1, e_2, \dots, e_n)$, the model autoregressively learns the conditional distribution:

$$P(E | MC) = \prod_{i=1}^n P(e_i | \hat{e}_{<i}, MC),$$

where $\hat{e}_{<i}$ represents the model's stochastic predictions for the preceding windows and MC denotes the mutational context (the observed data) across all windows. Concretely, the mutational context in a given window is the site frequency spectrum calculated across all samples, partitioned into the contributions from sites that are heterozygous and sites that are homozygous in the pivot pair.

At each genomic window i , the model outputs a probability distribution over discretized TMRCA bins:

$$\hat{e}_i \sim P_\theta(e_i | \hat{e}_{<i}, MC),$$

with P_θ parameterized by a neural network with weights θ . The training objective is to maximize the log probability of the true sequence given the (preceding) predicted sequence:

$$\mathcal{L}(\theta) = \sum_{i=1}^n \log P_\theta(e_i | \hat{e}_{<i}, MC),$$

where e_i denotes the ground truth label in window i , and $\hat{e}_{<i}$ are the model's own predictions up to (but not including) window i . Predictions are then centered around an intercept (log expected total diversity), which is sampled from the posterior distribution conditional on observed total diversity assuming a Poisson model of mutation. This objective encourages the model to learn a mapping from local mutation patterns to genealogical structure, enabling accurate sequential inference of coalescent events across the genome.

The right panel of Figure 1 illustrates **cxt** on a constant-size scenario ($N_e = 2 \times 10^4$, roughly equal mutation and recombination rates; see Datasets). All $\binom{50}{2}$ pivot pairs of 50 haploid chromosomes are inferred in parallel; for each pair the generative process is repeated 15 times, yielding samples from an approximate posterior over pairwise coalescence times along the sequence (where the prior is implicit via the simulated training set; see Calibration). Averaging across replicates produces stable estimates. In this well-specified setting **cxt** is highly accurate, and inference for all pairs completes in approximately five minutes on a single NVIDIA A100 GPU.

Benchmark Comparisons. To evaluate **cxt**'s performance, we benchmarked its accuracy against two approaches **Singer** and **SMC++** (11, 18), selected for their accuracy and broad adoption, respectively.

We first consider the performance of our “narrow model,” a **cxt** model trained on a constant-sized population ($N_e = 2 \times 10^4$) with equal mutation and recombination rates (see Datasets). In Figure 2, we aim to demonstrate two main results. First, we show the performance of the narrow model (**cxt-narrow**, top-left) on its original training domain, constant population size with roughly equal mutation and recombination rates, which yields a mean squared error (MSE) of 0.2531. As expected, **cxt-narrow**'s predictive accuracy declines when evaluated outside this domain. The top-middle panel shows results under a fluctuating population-size model (the sawtooth scenario; MSE 0.7496); thus, model misspecification degrades performance. We can potentially mitigate this behavior by training **cxt** on a wider range of parameters and demographic histories. To do so, we introduce a “broad model,” **cxt-broad** (top-right), trained on a substantially wider range of scenarios (see Datasets). **cxt-broad**, because it can accommodate demographic variation, achieves improved accuracy on the sawtooth model (MSE 0.1796).

Singer, combined with its dating-refinement algorithm **Polegon** (**Singer+Polegon**), attains comparable accuracy on the constant-size scenario (Figure 2 bottom-left; MSE 0.2470) and slightly reduced accuracy under the sawtooth model (MSE 0.2129). **SMC++**, which is primarily designed for demographic history inference with TMRCA decoding provided as part of its composite-likelihood framework, yields higher MSE values for these simulations (0.8685 and 1.7919), reflecting its emphasis on population-size reconstruction rather than fine-scale local genealogy prediction.

Because **cxt** is trained at a fixed sample size, users with larger samples can subsample, while for smaller samples we provide a lightweight adapter that fine-tunes the pretrained model to the target cohort size. SI Appendix, Fig. S4 shows adapter results transferring from $N = 25$ to $N = 5$ diploids on both the constant-size and sawtooth scenarios.

Our results demonstrate that the language modeling approach implemented in **cxt** is competitive with the most recent theory-driven methods, whether ARG-based (as in **Singer**) or SMC-based (as in **SMC++**), in these well-specified settings. A full summary of error estimates is provided in SI Appendix, Table S6 and Figure 2.

Next-coalescence prediction (Translation)

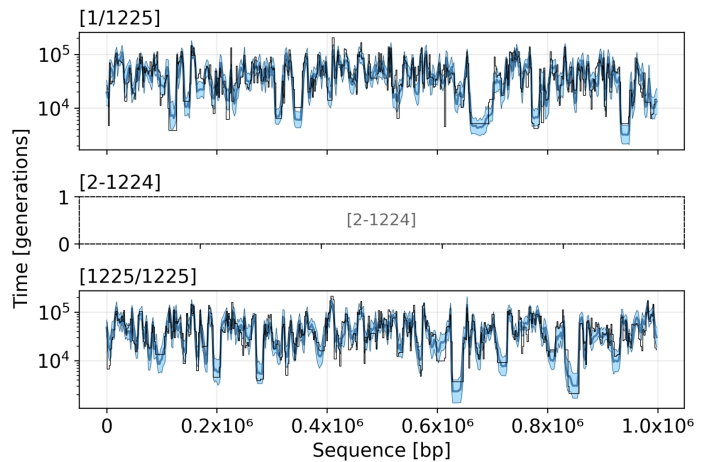
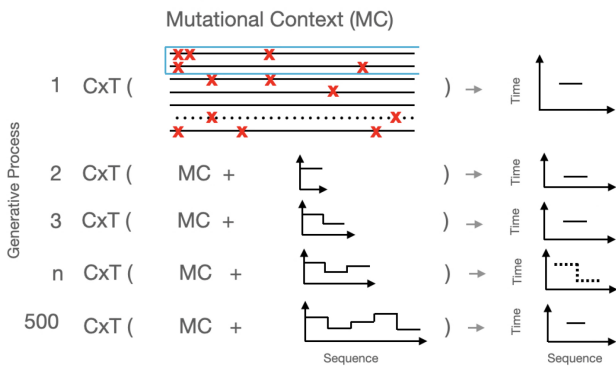


Fig. 1. *cxt* introduces the notation of *next-coalescence prediction* (left). *cxt* is a language model that conditions on a chosen “pivot” haplotype pair and predicts the pair’s time to the most recent common ancestor (TMRCA) for each window. *cxt* ingests mutational tensors constructed using the pivot pair and site–frequency–spectrum (SFS) values computed in windows across a focal region. The model works autoregressively: after each window is predicted, that estimate is appended to the context and supplied to the next step, yielding a step-wise reconstruction of the entire pairwise coalescent history. Because every haplotype pair can be processed in parallel, the approach scales efficiently with available GPU memory. In the example application (right), all $\binom{50}{2}$ pairwise coalescence curves for a sample of 50 haplotypes were inferred simultaneously in under five minutes on a single NVIDIA A100 GPU. True TMRCA are plotted in black, individual replicate predictions in light blue, and the prediction mean in dark blue.

Computational Efficiency and Scaling. We compared wall-clock runtimes of *cxt* with *SMC++* and *Singer* across increasing numbers of inferred pairwise coalescence trajectories (SI Appendix, Fig. S6). Overall, *cxt* exhibits approximately linear scaling in the number of inferred pairs and near-linear speedups with additional GPUs, while likelihood- and MCMC-based methods incur substantially higher and more parameter-sensitive computational costs. Detailed runtime breakdowns and scaling analyses are provided in the SI Appendix.

Towards a generalizable deep learning model. In the previous section, we focused primarily on the setting in which *cxt* is trained on a single demographic model (*cxt-narrow*), thereby learning directly from simulated data, while also providing a preview of *cxt-broad*’s potential. The central advantage of a language-modeling approach is its capacity, through many learnable parameters, to move beyond narrowly specified generative assumptions by conditioning its behavior on a rich set of mutational and genealogical contexts. In this sense, *cxt* functions as a flexible conditional density estimator: for a given context, it produces a corresponding predictive distribution. We assess this directly by comparing inferred marginal coalescence-time distributions across species.

In SI Appendix, Fig. S7, we report inferred coalescence time distributions across nearly all *stdpopsim* v0.2 species. For each species, we simulate under its published demographic model when available, and otherwise assume a constant population size. Results from simulations using fine-scale recombination maps are given separately in SI Appendix, Fig. S8. All results shown here are based on the *cxt-broad* model (see Datasets), which was trained across the full set of scenarios. Importantly, the evaluations in SI Appendix, Figs. S7 and S8 are performed on newly generated simulations that were not included in training, eliminating any opportunity for data leakage. Across nearly all species, the inferred distributions (blue) closely match the true distributions (black).

To promote generalization across demographic and rate regimes, we made a deliberate design choice during training: we keep the window size fixed at 2 kb while varying N_e over several orders of magnitude. In principle, one might expect the window size to scale with the effective population size, since the expected numbers of mutations and recombination events per window scale with N_e . However, by intentionally not adapting the window size, the model encounters windows that can be mutation-dense or mutation-sparse depending on the underlying N_e . This encourages the model to learn *relative* changes in local coalescent structure rather than relying on absolute mutation counts; any global offset can be corrected during inference by calibrating to the mutation rate.

A related consideration is how well the mean coalescent time within a window can be estimated. As the window size increases, estimation becomes easier: mutation counts rise, diversity increases, and sampling noise decreases. However, larger windows also average over spatial variation in genealogical structure, attenuating signals of population-size change, population structure, and other localized effects.

We illustrate this trade-off in SI Appendix, Fig. S5. Using 2 kb windows (left; MSE 0.1096) predicted coalescence times closely track the truth for the *stdpopsim* *AnoGam* model. In contrast, using much smaller 0.2 kb windows (middle; MSE 0.5727) substantially increases difficulty: more frequent recombination induces greater heterogeneity among local genealogies, while fewer mutations per window increase variance in the observed signal. Despite the broad model’s reasonable performance in this setting, we further fine-tune it on *stdpopsim* species with $N_e > 10^5$ (SI Appendix, Fig. S5, right). This yields the *broad* (*w200*) model and further reduces prediction error.

Out-of-sample tests for the broad model. During this project, *stdpopsim* v0.3 was released (2), expanding the catalog to include additional species (*S. scrofa*, *R. norvegicus*, *P. sinus*, *M. musculus*, and *G. gorilla*). We leveraged this release to assess the generalization ability of our broad

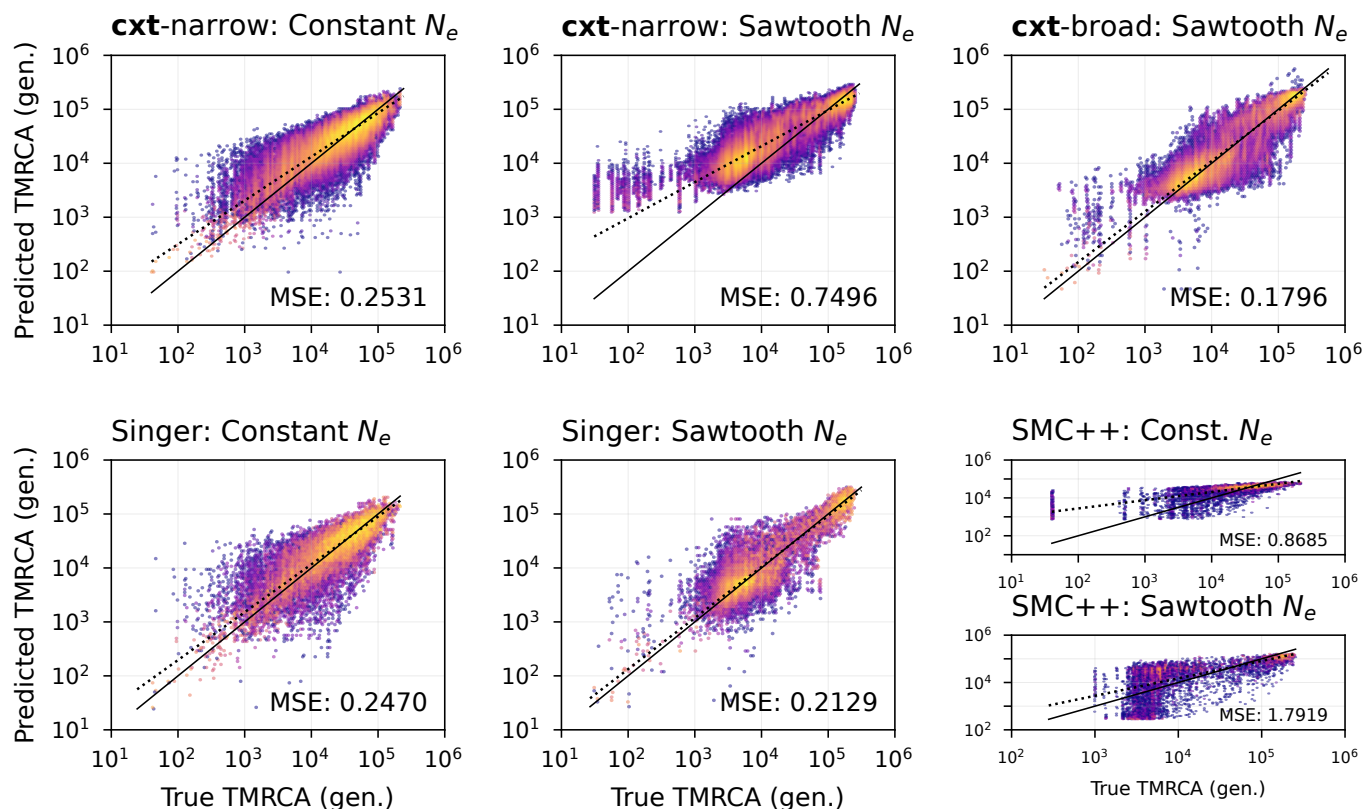


Fig. 2. True versus predicted coalescence times for three inference approaches across two demographic scenarios: a constant population size and a fluctuating “sawtooth” demography. Top row: *cxt*-narrow evaluated on the constant-size scenario (left) and on the sawtooth scenario (middle), followed by *cxt*-broad evaluated on the sawtooth scenario (right). The broad model—trained on a wider range of demographic histories—achieves substantially improved accuracy. Bottom row: *Singer*+*Polegon* evaluated on the constant-size (left) and sawtooth (middle) scenarios. The two rightmost panels show *SMC++* applied to the constant-size and sawtooth scenarios, respectively. Although *SMC++* is primarily designed for population-size inference, we include its decoding-based coalescence-time estimates for completeness. Mean squared errors (MSE) for each panel are reported within the plots.

model, testing performance *without* retraining. Because these species were not present in *stdpopsim* v0.2, they constitute genuinely out-of-sample test cases for the broad model.

The v0.3 release includes species spanning a wide range of effective population sizes, from very large populations, such as *S. scrofa* ($N_e = 270000$) and *R. norvegicus* ($N_e = 124000$), to much smaller populations, such as *P. sinus* ($N_e = 3500$). Most of these new species also exhibit recombination rates that exceed mutation rates. For example, in *P. sinus* we have $m = 5.83 \times 10^{-9}$ and $r = 1 \times 10^{-8}$ per bp per generation ($r/m = 1.715$). A notable exception is *G. gorilla* for which $r = 1.193 \times 10^{-8}$ and $m = 1.235 \times 10^{-8}$, yielding $r/m = 0.966$.

Results from this out-of-sample evaluation are shown in Figure 3. Notably, *cxt* generalizes well, even when applied to simulations generated under previously unseen regimes of recombination and mutation rates, population sizes and demographic structure. Across these v0.3 species, *cxt* generalizes reasonably well, although *Singer*+*Polegon* achieves lower error in most scenarios, in some cases by a substantial margin. Both methods substantially outperform the *SMC++* decodings.

Generalization across parameter regimes. A central goal of this project was to build a model that spans the diversity of scenarios represented in *stdpopsim*, motivated by the idea that as *stdpopsim* expands, we can incrementally incorporate new species into training. However, when applying

cxt to data from a new species, with potentially different mutation and recombination rates, effective population size or population structure, we expect inference accuracy to depend on how similar that species is to the regimes represented in the training data.

We assessed *cxt*’s ability to interpolate and extrapolate beyond the training distribution by evaluating prediction accuracy across a grid of mutation and recombination rates at fixed $N_e = 20,000$ (SI Appendix, Fig. S10). The grid is chosen to span most *stdpopsim* species; MSE and KL divergence heatmaps show that, as expected, error is highest in the low-mutation, high-recombination regime where the signal-to-noise ratio is least favorable. Representative TMRCA predictions from the four corners of the grid (SI Appendix, Fig. S11) confirm that *cxt* generalizes beyond the parameter regions represented in training.

Calibration of approximate posteriors. In addition to point estimates, *cxt* produces approximate posterior samples for local TMRCA, enabling uncertainty quantification. We evaluated the frequentist calibration and consistency of these posteriors across multiple demographic scenarios and mutation-rate regimes, including out-of-training species. Overall, posterior intervals show near-nominal empirical coverage and contract appropriately with increasing mutational information, with deviations confined to boundary windows where context is truncated. Detailed calibration

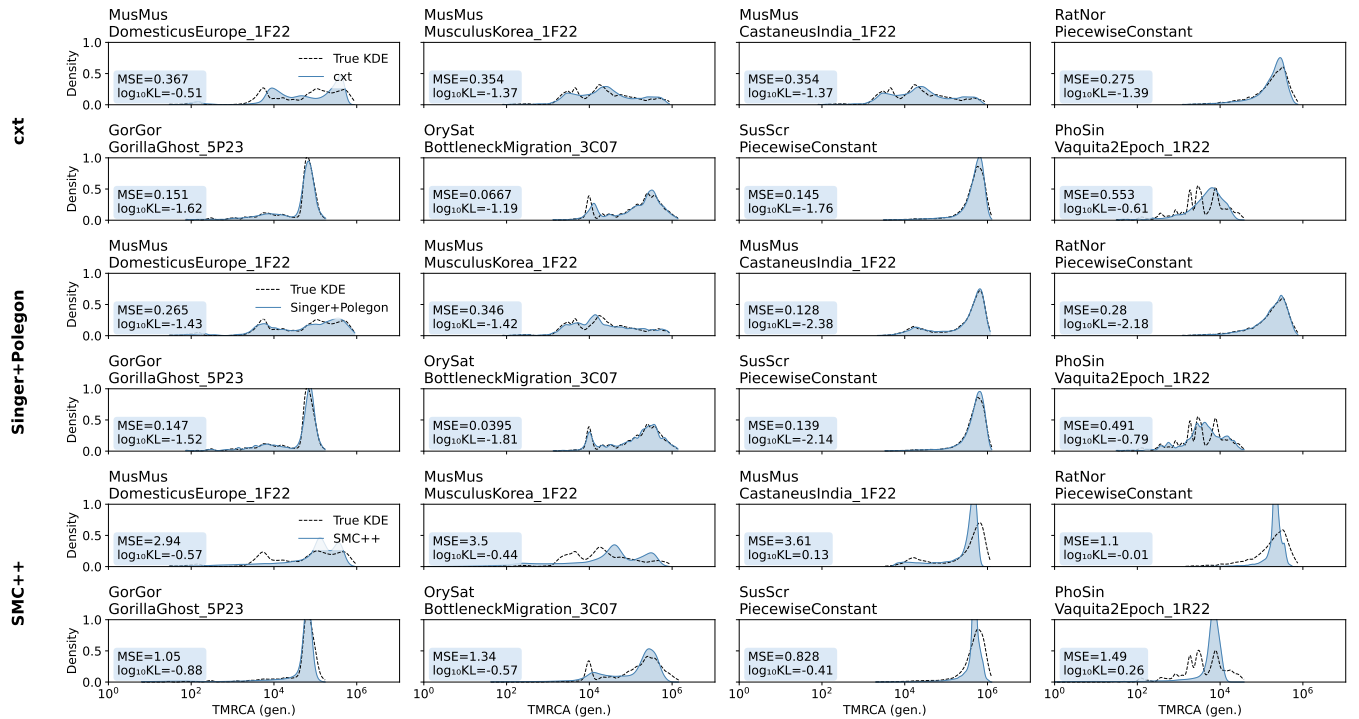


Fig. 3. Out-of-sample evaluation of the broad model on stdpopsim v0.3. Each panel shows inferred marginal coalescence distributions (dashed) against true distributions (shaded). We simulated data from the novel species / demographic histories added in stdpopsim v0.3, aggregating pairwise inferences into kernel density plots. All results are from simulations previously unseen during training (Top Rows: cxt; Middle Rows: Singer+Polegon; Bottom Rows: SMC++).

analyses are provided in SI Appendix, Figs. S2, S3, and S9.

Demography estimation through coalescence rates. Estimating pairwise coalescence rates provides a direct window into changes in effective population size through time, perhaps most famously exemplified by the original PSMC model of Li and Durbin (2011). To evaluate our ability to recover effective population size trajectories from cxt predictions, we simulated data under demographic models for *H. sapiens*, *B. taurus*, and *A. thaliana*. From these simulations, we estimated pairwise coalescence-time distributions and converted them into coalescence-rate curves, which we then mapped to estimates of historical effective population size (Figure 4; see Methods).

Figure 4 shows five curves. The black *expectation* line reflects the true coalescence rate implied by the generative model. We also report an *inference limit* in red, constructed by sampling the true TMRCA every 2 kb across the sequence and then estimating coalescence rates using the same procedure applied to cxt, Singer, and SMC++, thereby providing an upper bound under matched discretization. The remaining curves show the corresponding estimates from cxt, Singer, and SMC++.

Overall, effective population size trajectories inferred from cxt and Singer closely track the truth at a genome-wide scale capturing the major epochs of growth/bottlenecks, whereas SMC++ shows larger deviations at recent times; this gap may reflect limited information in the current setting and could plausibly be reduced by incorporating additional data.

We also inferred the cross-coalescence rate (i.e., the rate of coalescence between lineages drawn from different demes) under the two-population OutOfAfrica_2T12

model from stdpopsim. Generally, cross-coalescence curves are informative about the timing of divergence and subsequent gene flow between populations within a species. SI Appendix, Fig. S12 summarizes these results. The left panel shows the cross-population rate estimated from one African and one European haplotype, whereas the middle and right panels show the within-population rates for African and European samples, respectively. The cross-population curve (left) closely follows the expected trajectory. In contrast, the within-population curves exhibit a modest deviation for cxt.

A likely explanation is the way structured models were represented during training: for each simulated instance, 50 samples were drawn with random population assignments. As a result, training examples rarely contained only a single population, which may bias within-population rate estimation. This could potentially be mitigated by modifying the training scheme for structured models, for example by explicitly including single-deme sampling configurations.

These demographic inferences are based on 10 Mb of simulated sequence (Figure 4 and SI Appendix, Fig. S12); because shorter sequences truncate the longest tracts of recent ancestry, increasing variance in recent-time rate estimates, we expect substantially improved accuracy at full chromosome lengths. Overall, cxt's local TMRCA predictions aggregate into accurate genome-scale demographic summaries.

Application to Human 1000 Genomes and Ag1000G Mosquito data. Having established the accuracy and robustness of cxt across a broad range of simulated scenarios, we next apply it to empirical genomic data: human variation from the

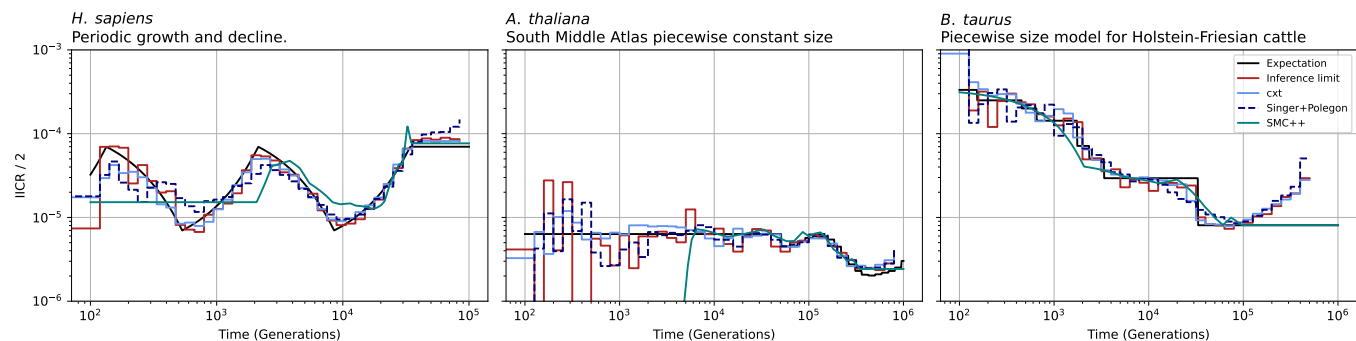


Fig. 4. Inverse-instantaneous coalescence rate calculation for the piecewise-constant demography of *H. sapiens* (left), *A. thaliana* (middle) and *B. taurus* (right). The inference of pairwise-coalescence events leads to the implicit inference of demography estimates through the marginal coalescence distribution assuming coalescence occurs as a Poisson process (see Materials and Methods). For each scenario 10 Mb and 25 diploid samples have been used to achieve resolution throughout the specified time-windows.

1000 Genomes Project (47) and mosquito genomes from the Ag1000G consortium (49).

We begin with the human dataset, revisiting the canonical selective sweep at *LCT* as a validation benchmark, and then turning to more recently dated regions of the *HLA* locus. In the latter case, our inferred TMRCA patterns are broadly consistent with previously reported old genealogical ages, though we infer slightly younger times than *Singer+Polegon*.

We then analyze the insecticide-resistance loci *Rdl* in *Anopheles gambiae*. This locus has been highlighted using summary-statistic approaches, including Garud's H_1/H_{12} , but a clear time-resolved interpretation of the underlying sweep dynamics has remained elusive. Using *cxt*, we estimate fine-scale coalescent times across five African populations (Burkina Faso, Mali, Cameroon, Ghana, and Uganda) while addressing three technical challenges characteristic of Ag1000G data, and we compare our mosquito estimates to *Singer+Polegon*:

1. **Large effective population sizes**, which yield deep genealogies and correspondingly large scaled mutation and recombination rates;
2. **Small sample sizes** in Ghana (< 25 diploid individuals); and
3. **Substantial missing-data variation** across genomic regions

To begin this initial validation in humans, we selected 50 individuals of British descent (GBR), and estimated coalescent times for 25 focal pairs, holding out the remaining $n = 48$ samples as the AFS input to *cxt*.

We choose 25 focal pairs because this sample size allows all chromosome-wide comparisons to fit in a single batch on an A100 GPU, enabling rapid whole-chromosome inference. The top panels of Figure 5 show chromosome-wide TMRCA landscapes for chromosomes 2 and 6. As an additional validation, we compared inferred TMRCA against the number of recombination events predicted by *cxt* (SI Appendix, Fig. S15), recovering the expected inverse relationship between genealogical age and segment length.

The bottom panels in Figure 5 zoom in on two canonical regions: the *LCT* locus on chromosome 2 and the *HLA* region on chromosome 6. At *LCT*, we recover markedly reduced coalescent times as expected under the well-characterized recent selective sweep at this locus. *cxt*

captures this local trough cleanly. Around *LCT*, estimated coalescent times slightly above 10 kya are consistent with the origin of the sweeping haplotype, and the variation across focal pairs highlights local heterogeneity.

The *HLA* region provides the complementary pattern: exceptionally deep genealogical structure shaped by long-term balancing selection. The extended peaks inferred by *cxt* (bottom-right) match long-standing expectations for this locus and support the conclusion that the model captures extreme TMRCA variation. Consistent with this, we replicate the multi-peaked ancient structure reported by (51), with several genes reaching TMRCA on the order of tens of millions of years, one of the strongest known signatures of long-term balancing selection.

To investigate selective dynamics at the *Rdl* locus across multiple *A. gambiae* populations, we constructed a comparison panel (SI Appendix, Fig. S14) showing coalescent-time landscapes from *cxt*, *Singer+Polegon*, and *SMC++*. The challenges outlined above are particularly acute here; high N_e -scaled rates substantially slow MCMC-based approaches such as *Singer+Polegon*. For readability, we show only *cxt* results in the main text (Figure 6; see also SI Appendix, Fig. S13); full method comparisons appear in SI Appendix, Fig. S14.

To address these issues, we fine-tuned *cxt* on a filtered *stdpopsim* training set restricted to $N_e > 10^5$, reduced the window size ten-fold to 200 bp, and performed inference in 100 kb blocks. We additionally incorporated Ag1000G missing-data masks during fine-tuning to improve robustness to region-specific missingness.

Finally, because the Ghana population does not meet the 25-diploid sample size used elsewhere, we trained a small adapter model to handle reduced sample size, while jointly accounting for large N_e and missing-data patterns.

The *Singer+Polegon* method was also run on the same samples. We tried two approaches to using *Singer+Polegon*: one with the “natural” parameterization of mutation and recombination rates, which for *Anopheles* yields a ratio of recombination:mutation $\gg 1$, and a second where we held this ratio fixed at 1 (equivalent to the human parameterization). The former, “natural” parameterization yielded poor fits to observed diversity, particularly the marginal site frequency spectrum, thus the results presented here do not use mutation and recombination rates that reflect what is known about the *Anopheles* genome, although the resulting fits are superior with respect to matching observed diversity patterns.

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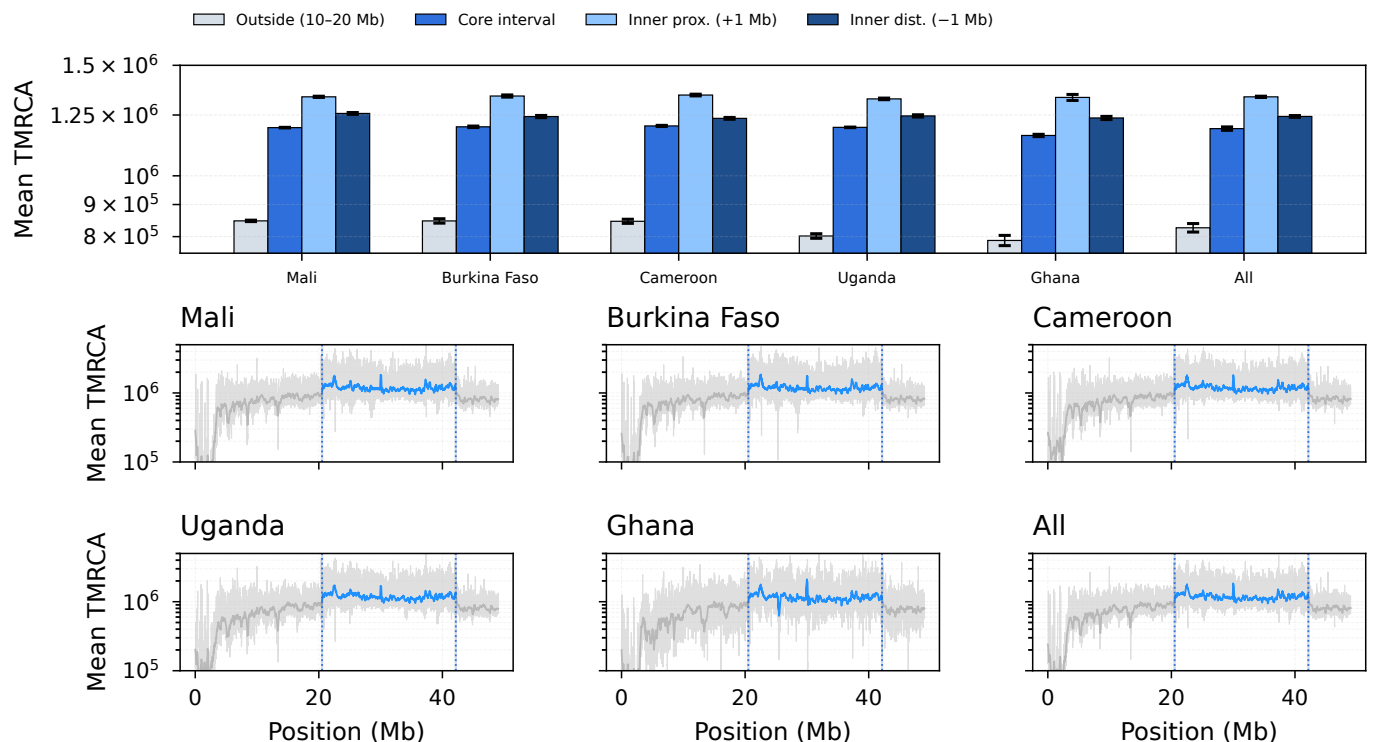


Fig. 6. Coalescent-time structure across the *In(2L)a* inversion on chr2L. Top: mean TMRCA summaries for an outside background region (10–20 Mb), the full inversion core interval, and two interior 0.5 Mb windows positioned 1 Mb inside each breakpoint (mean $\pm$ s.e.m. across replicate trajectories; “All” aggregates populations). Bottom: genome-wide mean TMRCA curves shown per population and pooled (“All”), with vertical guides marking inversion boundaries, breakpoint-adjacent windows, and the interior sampling windows used for summary statistics.

in parallel. This approach differs fundamentally from prior deep learning methods such as ReLERNN (41) or CoalNN (40), which treat inference along the chromosome as a supervised regression task, and produce a single point estimate from sequence data. In contrast, *cxt* uses a language model to learn the underlying conditional stochastic process.

cxt produces genome-wide TMRCA estimates for a diploid individual in minutes on a single GPU, and it can alternatively focus on a specific region and decode many pairs locally. Comparisons to SMC++ and Singer+Polegon show that *cxt* consistently outperforms SMC++ in terms of accuracy of TMRCA inference (SI Appendix, Table S6), and performs competitively with the state-of-the-art ARG-based method Singer+Polegon, matching it in well-specified settings, though with higher error in some out-of-distribution scenarios. Singer+Polegon remains the more accurate choice when computational cost is not a constraint, but achieving competitive performance without explicit ARG modeling highlights the efficiency of the language-model approach. These computational and statistical properties motivate application to empirical datasets spanning both canonical and challenging inference regimes.

To assess the performance of *cxt* on empirical data, we applied it to two empirical systems with very different mutation-to-recombination regimes: (i) a targeted validation in humans of the *HLA* and *LCT* regions of the genome, known coalescent time outliers, and (ii) a chromosome-scale scan of chr2L in *Anopheles*, highlighting the extensibility of our model to settings with huge

population sizes and pervasive missing data that challenge standard statistical estimation.

In humans, the *LCT* locus provides a canonical example of recent, strong, and geographically structured selection. Lactase persistence is driven primarily by regulatory variation upstream of *LCT*, including variants located in intronic sequence of the neighboring *MCM6* gene that modulate enhancer/promoter activity (53, 54). Population-genetic analyses indicate that derived lactase-persistence haplotypes rose rapidly in frequency over roughly the past 5,000–10,000 years, consistent with gene-culture coevolution in the context of dairying (55, 56). Our pairwise TMRCA estimates at *LCT* reflect this known history cleanly, recovering a pronounced local dip in coalescence times centered on the locus. In our sample, some pairs exhibit a TMRCA younger than 10,000 years, substantially younger than the average across the region, but we note that there is extensive variation across pairs that likely reflects that *LCT* arose from standing variation (57), and thus the locus harbors significant variability in the underlying genealogies.

In the *HLA* region, we observe exceptionally ancient separation times among alleles. For several genes, pairwise TMRCA exceed commonly cited human–chimpanzee divergence times, consistent with trans-species polymorphism under long-term balancing selection. The *HLA* locus is among the most polymorphic regions of the human genome, likely driven to extreme variation by arms-race dynamics at the immune genes it encodes. Accordingly, there is abundant evidence that some *HLA* alleles have been maintained for tens of millions of years by balancing selection (58–60). In our sample of British individuals,

the *HLA* region is a clear outlier in the chromosome-wide distribution of coalescence times (Figure 5, top right). Across multiple genes, we observe pairwise TMRCA exceeding 10 million years. While surprisingly ancient, these estimates are nonetheless in agreement with recent population-genetic dating in (51). Together, *LCT* and *HLA* provide complementary extremes, recent directional selection versus long-term balancing selection, allowing us to validate *cxt* across nearly the full dynamic range of human genealogical timescales, from thousands to tens of millions of years.

Unlike the human analyses, which primarily serve as validation against well-characterized loci, the Ag1000G application reveals previously unresolved spatial and temporal structure within ancient inversions and insecticide-resistance sweeps. These mosquito data also present a substantially more challenging setting for coalescent inference. In particular, the Ag1000G data combine very large effective population sizes, pervasive and spatially heterogeneous missingness, and uneven sample sizes across populations. Together, these factors challenge both diversity-based summaries and likelihood-based decoding, and provide a stringent test of whether *cxt* can be adapted to real data with non-ideal sampling and observation patterns.

We highlight chromosome 2L which carries a number of known targets of insecticide resistance as well as a well studied inversion, *In(2L)a*. Considering first the chromosomal landscape of estimates (SI Appendix, Fig. S13, bottom panels), we observe that the *In(2L)a* inversion region displays generally deeper coalescent times than the surrounding genomic background, with particular peaks in age near its breakpoints; consistent with prior observations of elevated diversity within this inversion (61). Previous age estimates of the inversion using phylogenomics have aged the origin of the inversion to more than two million years ago (62), before the origin of the *An. gambiae* complex of species. We find that the regions proximal to the inversion breakpoints show ages on the order of ~ 1.1 million generations, consistent with this deep origin. This pattern likely reflects a combination of suppressed recombination between inverted and standard arrangements, as well as long-term balancing selection maintaining both arrangements in the population. While the deep origin of *In(2L)a* is well established, how such an ancient inversion has been maintained within populations, and whether its evolutionary history is uniform across its length, remains less well understood. The pronounced and spatially structured coalescent age peaks near the inversion breakpoints indicate that different regions of the inversion have experienced markedly different genealogical histories since its origin. This suggests that recombination suppression and long-term maintenance have acted heterogeneously across the inversion. More broadly, it demonstrates that population-scale genealogical inference can reveal biologically meaningful structure in the evolutionary dynamics of ancient chromosomal inversions beyond a single global age estimate.

Along chromosome 2L, a significant outlier is the well-studied *Rdl* locus, which confers resistance to dieldrin and other cyclodiene insecticides (63–65). Resistance to dieldrin appeared in Nigerian populations only ~ 18 months after the initial onset of insecticide control efforts in 1954 (66) as a single, dominant allele (67). Population

genetic evidence points to a strong selective sweep at *Rdl* as well as a complex history of introgression that extends between species and inversion karyotypes (52). Our coalescent time estimates fill in this picture; we find clear evidence of recent selection at the locus, with short coalescent times among individuals in populations, such as Ghana, in which the resistant *Rdl* is at high frequency. Conversely in east African samples, where resistance is not found, coalescent times at the locus are indistinguishable from the local background.

Generally, our youngest time estimates at *Rdl* are on the order of hundreds to thousands of years, which would predate the introduction of broad insecticide use. Two things could be pushing back our dates here: i) it could be that the mutation was segregating at low frequency before the onset of selection, and ii) our estimates average over sites, and are not dating the origin of the *A296G* nonsynonymous mutation that confers resistance in *An. gambiae*.

Taken together, these applications show that *cxt* remains effective on large empirical datasets, including local genomic regions that deviate substantially from the evolutionary scenarios represented in its simulation-based pretraining, which acts as an implicit prior, such as targets of selective sweeps and long-term balancing selection. The Ag1000G analyses, in particular, highlight a practical advantage of simulation-based models: the ability to adapt to new evolutionary regimes with minimal targeted modification. By fine-tuning on high- N_e simulations, reducing the window size to 200 bp, incorporating empirical missing-data masks, and adding a lightweight adapter for small sample sizes, *cxt* remained accurate in settings where traditional ARG MCMC and SMC-based methods face computational or modeling limitations. This flexibility, combined with amortized inference, positions *cxt* as a practical and scalable tool for population-genomics studies as community resources such as *stdpopsim* continue to expand to encompass additional species, demographic histories, and recombination landscapes.

In summary, we introduce a language-model-based inference engine designed to scale with the growing *stdpopsim* catalog. By directly learning a conditional stochastic process, *cxt* absorbs prior knowledge directly through simulation-based training, without requiring new analytic likelihood derivations for each scenario. As a result, *cxt* is a fast and accurate tool for TMRCA inference, and provides a flexible framework that can be adapted to novel evolutionary settings.

Materials and Methods

Next-coalescence Prediction. We introduce as our prediction task, *next-coalescence* prediction, which we define to mean the next point in time in which a focal (or pivot) pair of sequences find common ancestry as we move across (say from left to right) a sequence. Our method processes 500 genome windows (most commonly e.g. 500×2 Kb or 1Mb; or 500×200 bp or 0.1 Mb) of sequence. For the purposes of this paper we use a sample size of 25 diploid individuals, but this is arbitrary and we provide a method to train a light-weight adapter generalizing to required input sizes (or recommend sub-sampling and averaging for larger sample sizes). Our model focuses on a pivot pair of chromosomes, also non-as distinguished pairs in SMC++ terminology, from which we predict pairwise coalescence events, or more specifically a probability distribution from which the next coalescence event can be sampled for that pivot pair along the sequence for each of the 500 windows. This describes an iterative process by which the predicted coalescence time is concatenated with

the mutational context, forming the next input to the model to predict the next coalescence event (see Figure 1). We refer to this procedure as *next-coalescence prediction*. We continue by describing the processing of the genotype matrix as obtained from coalescent simulation before presenting the architecture and listing the specific simulation parameterization next.

Processing of Coalescent Simulations. We begin with a genotype matrix $G \in \mathbb{Z}^{2N \times M}$ containing $2N$ haploid sequences (from N diploid individuals) across M sites, obtained from coalescent simulations (here `msprime` (68)) or empirical data. All empirical analyses in this work use within-individual pivot pairs (the two haplotypes of one diploid, analogous to the PSMC setting), for which phasing is not required: heterozygous and homozygous sites are determined by the genotype alone, so the XOR/XNOR features and SFS weights that constitute the model input are identical whether or not phase is known. Cross-individual pivot pairs, used here only in simulated benchmarks where phase is known by construction, would require phased haplotypes; in that setting, standard statistical phasing tools can be applied as a routine upstream step.

A high-level description is provided here, with an explicit version in SI Appendix, Algorithm 1 and a schematic overview in SI Appendix, Fig. S1. The core preprocessing consists of extracting mutation patterns for each pivot pair using logical operations—heterogeneous differences via XOR and homogeneous matches via XNOR—and then weights these pivot-pair vectors by their site-frequency spectrum (SFS) counts computed across all remaining samples. This converts the otherwise binary mutation indicators from the logical operations into coarse age-informed signals, ensuring that the inferred pairwise coalescence times are driven by the allele frequency spectrum present in the entire sample.

This procedure is applied in sliding windows of increasing size (4–128 Kb) with a 2 Kb stride, which allows for rescaling when inference is performed at different resolutions.

In summary, the model input consists solely of the genotype matrix and variant positions; from these, we derive windowed, SFS-weighted mutation patterns for each pivot pair, which the language model uses to estimate average coalescence times per window. These inputs arise naturally from `msprime` simulations or from empirical genotype data such as VCFs.

Implementation of the Decoder-Only Architecture. We implement a decoder-only transformer closely aligned with GPT-2 (69), rather than the original encoder–decoder formulation (70), and adapt it to the structure of population genetic data. In NLP, text is tokenized and mapped to embeddings via a learned lookup table; in contrast, the mutation patterns in genomic windows form a vast, effectively continuous space that makes discretization impractical. Instead of defining a bespoke tokenizer, we directly project SFS-weighted mutation patterns into the latent space using a fully connected embedding module (SI Appendix, Algorithm 2), which accepts tensors of shape $B \times Z \times WS \times K \times N$ and outputs $B \times K \times E$. Here, B is batch size, Z the logical vector distinguishing heterozygous and homozygous states, WS the number of window sizes, K the number of windows, N the number of samples, and E the embedding dimension.

Coalescence times are represented by partitioning log-scaled values into 324 discrete intervals (SI Appendix, Table S1), each mapped through a small embedding table. The model then concatenates mutation and coalescence embeddings so that every context window carries both the local mutational signal and the discretized temporal state. This setup allows the same model to perform inference across diverse evolutionary scenarios without re-training, since the inputs already encode the relevant population-genetic context.

To encode genomic position, we adopt rotary positional embeddings (71), which efficiently capture spatial structure and require no adjustment under fixed-stride windows (e.g., 2 kb). The resulting sequence fed to the transformer consists of 500 mutation embeddings, a start token, and the coalescence embeddings, yielding a fixed latent length of 1001; padding exists but is unused. A dropout rate of 0.1 is applied for regularization. Both narrow (single-scenario) and broad (multi-scenario) models (`cxt-narrow` and `cxt-broad` respectively) share this architecture, with parameterization summarized in SI Appendix, Table S1. Further implementation details are available at <https://github.com/kr-colab/cxt>.

Sample-size and accessibility adaptation. `cxt` models are trained at a fixed sample size, which determines the number of SFS channels used to represent the mutational context. Empirical datasets may contain fewer samples than the pretrained model (e.g. the Ghana population in Ag1000G), and retraining the full model for each sample size would be inefficient. To enable transfer to smaller cohorts, we introduce a lightweight input-adaptation stage that operates before the pretrained decoder-only transformer.

Sample-size adapter. The adapter maps SFS features computed from a reduced cohort to the dimensionality expected by the pretrained model. After this transformation, data are processed by the original projection layers and transformer without further modification. In this way, the adapter modifies only how mutational information is presented to the model, while preserving the pretrained representation of the coalescent process.

The adapter is implemented as a small residual neural network applied independently to each genomic window. It consists of layer normalization followed by a gated bottleneck that rescales and recombines SFS channels derived from the smaller cohort. A residual connection projects the input directly to the target dimensionality, and a learnable scaling parameter initialized near zero controls the magnitude of the adapter's contribution. As a result, the adapter behaves close to an identity mapping at the start of training and fine-tunes in a stable manner.

Training procedure. Adapter training is performed by freezing all parameters of the pretrained decoder-only transformer and updating only the adapter parameters. The training objective is identical to that of the full model, namely prediction of the local pairwise coalescence time in each genomic window. Because the adapter contains far fewer parameters than the backbone, it converges rapidly and can be trained in a small number of epochs using simulations generated at the target sample size.

To accommodate small but systematic shifts in input statistics, we additionally unfreeze the LayerNorm parameters and the final transformer blocks and train them at a substantially reduced learning rate. This restricted fine-tuning improves stability when transferring to smaller cohorts while avoiding degradation of the pretrained coalescent prior.

Fine-tuning to genomic inaccessibility. Empirical genomes contain extended regions that are systematically uncalled, producing structured missingness patterns that differ from idealized simulations. To account for this, we fine-tune the adapted model on training examples in which genomic inaccessibility is imposed explicitly using an empirical accessibility mask.

For each simulated tree sequence, we select a contiguous genomic segment and delete intervals corresponding to inaccessible sites from the tree sequence, ensuring that both mutational and genealogical information reflect the empirical observation process. In parallel, we compute the fraction of inaccessible bases per window across multiple scales and inject this signal into a dedicated input channel, providing the model with explicit information about local callability. This fine-tuning step uses the same training objective as above and updates the same restricted set of parameters (adapter, LayerNorms, and final transformer blocks), allowing the model to retain its pretrained coalescent representation while calibrating to irregular observation patterns in real genomes.

Calibration of the Molecular Clock. Although `cxt` is mutation-rate agnostic during training, its raw TMRCA predictions must be placed on an absolute timescale for demographic interpretation. We therefore apply a simple calibration step that rescales predicted TMRCA so that the expected nucleotide diversity matches the observed diversity across the full context window. This correction is typically negligible when the data resemble the training scenarios, but becomes useful under atypical mutation rates or demographic histories. To avoid collapsing variability across replicates, we treat the correction factor probabilistically rather than deterministically. A full derivation, including the Poisson–Gamma model used to sample per-replicate correction factors, is provided in SI Appendix, Figs. S2 and S3.

Estimating Population Size from Coalescence Rates. To convert predicted TMRCA into population size trajectories, we estimate the instantaneous coalescence rate on a logarithmic time grid and invert it to obtain $N_e(t) = 1/(2\lambda(t))$. This provides a nonparametric demographic reconstruction directly from the model's TMRCA samples. Details, including windowing, discretization, and treatment of the tail interval, appear in SI Appendix.

Datasets and Training Description. Full details of the training/validation parameter sets (narrow vs. broad; `stdpopsim` v0.2 for training and v0.3 reserved for validation), along with exact commands, are provided in SI Appendix and in the online manual. Models were implemented in PyTorch Lightning and trained with AdamW (base learning rate 3×10^{-4}) with cosine annealing; datasets were stored in float16 for efficiency. Fine-tuned variants were trained for two epochs with a ten-fold reduced learning rate. Inputs follow a fixed-length mutation context paired with coalescence tokens, trained with a causal NLL objective. We use a translation-aware fused-causal mask so coalescence predictions attend to all prior mutations and past coalescences, while mutation predictions attend to all prior tokens.

Environmental Considerations. Although GPUs typically draw more instantaneous power than CPUs (e.g., ~250–400 W for an NVIDIA A100 versus ~225 W for a high-end server CPU), they substantially reduce wall-clock time for the parallel inference workloads used by cxt (training requires on the order of 10 h on three A100s). Because amortized inference is insensitive to recombination-rate variation, GPU acceleration can reduce total energy use and CO₂ emissions relative to CPU-based likelihood methods, particularly in large-*N_e* regimes.

Data, Materials, and Software Availability. Simulation code, model training workflows, and accompanying documentation are available under a CC-BY-NC 4.0 license at <https://github.com/kr-colab/cxt>, with additional resources at <https://cxt.readthedocs.io/en/latest/>. Publicly available data from the 1000 Genomes Project were used for evaluation.

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1. J. R. Adion, et al., A community-maintained standard library of population genetic models. *eLife* **9**, e54967 (2020).
2. M. E. Lauterbur, et al., Expanding the stdpopsim species catalog, and lessons learned for realistic genome simulations. *eLife* **12**, e84874 (2023).
3. G. Gower, N. S. Pope, M. F. Rodrigues, et al., Accessible, realistic genome simulation with selection using stdpopsim. *bioRxiv* (2025).
4. J. Kingman, The coalescent. *Stoch. Process. Appl.* **13**, 235–248 (1982).
5. R. R. Hudson, Properties of a neutral allele model with intragenic recombination. *Theor. Popul. Biol.* **23**, 183–201 (1983).
6. R. C. Griffiths, P. Marjoram, An ancestral recombination graph in *Progress in population genetics and human evolution*. pp. 257–270 (1997).
7. R. C. Griffiths, The two-locus ancestral graph in *Institute of Mathematical Statistics Lecture Notes - Monograph Series*. pp. 100–117 (1991).
8. J. Kelleher, et al., Inferring whole-genome histories in large population datasets. *Nat. Genet.* **51**, 1330–1338 (2019).
9. L. Speidel, M. Forest, S. Shi, S. R. Myers, A method for genome-wide genealogy estimation for thousands of samples. *Nat. Genet.* **51**, 1321–1329 (2019).
10. B. C. Zhang, A. Biddanda, Å. F. Gunnarsson, F. Cooper, P. F. Palamara, Biobank-scale inference of ancestral recombination graphs enables genealogical analysis of complex traits. *Nat. Genet.* **55**, 768–776 (2023).
11. Y. Deng, R. Nielsen, Y. S. Song, Robust and accurate Bayesian inference of genome-wide genealogies for hundreds of genomes. *Nat. Genet.* **57**, 2124–2135 (2025).
12. R. Griffiths, P. Marjoram, Ancestral inference from samples of DNA sequences with recombination. *J. Comput. Biol.* **3**, 479–502 (1996).
13. C. Wiuf, J. Hein, Recombination as a point process along sequences. *Theor. Popul. Biol.* **55**, 248–259 (1999).
14. G. A. McVean, N. J. Cardin, Approximating the coalescent with recombination. *Philos. Trans. R. Soc. B* **360**, 1387–1393 (2005).
15. H. Li, R. Durbin, Inference of human population history from individual whole-genome sequences. *Nature* **475**, 493–496 (2011).
16. S. Schiffels, R. Durbin, Inferring human population size and separation history from multiple genome sequences. *Nat. Genet.* **46**, 919–925 (2014).
17. A. Malaspina, et al., A genomic history of Aboriginal Australia. *Nature* **538**, 207–214 (2016).
18. J. Terhorst, J. A. Kamm, Y. S. Song, Robust and scalable inference of population history from hundreds of unphased whole-genomes. *Nat. Genet.* **49**, 303–309 (2017).
19. T. P. P. Sellinger, D. A. Awad, M. Moest, A. Tellier, Inference of past demography, dormancy and self-fertilization rates from whole genome sequence data. *PLoS Genet.* **16**, e1008698 (2020).
20. K. Korfmann, T. P. P. Sellinger, F. Freund, M. Fumagalli, A. Tellier, Simultaneous inference of past demography and selection from the ancestral recombination graph under the beta coalescent. *Peer Community J.* **4** (2024).
21. T. Sellinger, F. Johannes, A. Tellier, Improved inference of population histories by integrating genomic and epigenomic data. *eLife* **12** (2024).
22. P. F. Palamara, J. Terhorst, Y. S. Song, A. L. Price, High-throughput inference of pairwise coalescence times identifies signals of selection and enriched disease heritability. *Nat. Genet.* **50**, 1311–1317 (2018).
23. R. Schweiger, R. Durbin, Ultra-fast genome-wide inference of pairwise coalescence times (2023).
24. J. Terhorst, Accelerated Bayesian inference of population size history from recombining sequence data. *bioRxiv* p. 2024.03.25.586640 (2024).
25. M. A. Beaumont, W. Zhang, D. J. Balding, Approximate Bayesian computation in population genetics. *Genetics* **162**, 2025–2035 (2002).
26. S. Boitard, W. Rodríguez, F. Jay, S. Mona, F. Austerlitz, Inferring population size history from large samples of genome-wide molecular data - an approximate Bayesian computation approach. *PLoS Genet.* **12**, e1005877 (2016).
27. F. Jay, S. Boitard, F. Austerlitz, An ABC method for whole-genome sequence data: Inferring Paleolithic and Neolithic human expansions. *Mol. Biol. Evol.* **36**, 1565–1579 (2019).
28. J. Min, Y. Ning, N. S. Pope, F. Baumdicker, A. D. Kern, Neural posterior estimation for population genetics. *bioRxiv* pp. 2025–12 (2025).
29. J. Schmidhuber, Annotated history of modern AI and deep learning (2022).

30. K. Korfmann, O. E. Gaggiotti, M. Fumagalli, Deep learning in population genetics. *Genome Biol. Evol.* **15**, evad008 (2023).
31. S. Sheehan, Y. S. Song, Deep learning for population genetic inference. *PLoS Comput. Biol.* **12**, e1004845 (2016).
32. C. Battey, P. L. Ralph, A. D. Kern, Predicting geographic location from genetic variation with deep neural networks. *eLife* **9**, e54507 (2020).
33. A. D. Kern, D. R. Schrider, diploS/HIC: an updated approach to classifying selective sweeps. *G3 (Bethesda)* **8**, 1959–1970 (2018).
34. L. Flagel, Y. Brandvain, D. R. Schrider, The unreasonable effectiveness of convolutional neural networks in population genetic inference. *Mol. Biol. Evol.* **36**, 220–238 (2019).
35. L. Torada, et al., lmaGene: a convolutional neural network to quantify natural selection from genomic data. *BMC Bioinforma.* **20**, 337 (2019).
36. T. Sanchez, J. Cury, G. Charpiat, F. Jay, Deep learning for population size history inference: Design, comparison and combination with approximate Bayesian computation. *Mol. Ecol. Resour.* **21**, 2645–2660 (2021).
37. L. S. Whitehouse, D. R. Schrider, Timesweeper: accurately identifying selective sweeps using population genomic time series. *Genetics* **224**, iyad084 (2023).
38. C. C. Smith, S. Tittes, P. L. Ralph, A. D. Kern, Dispersal inference from population genetic variation using a convolutional neural network. *Genetics* **224**, iyad068 (2023).
39. C. C. Smith, A. D. Kern, disperseNN2: a neural network for estimating dispersal distance from georeferenced polymorphism data. *BMC Bioinforma.* **24**, 385 (2023).
40. J. N. Saada, Z. Tsangalidou, M. Stricker, P. F. Palamara, Inference of coalescence times and variant ages using convolutional neural networks. *Mol. Biol. Evol.* **40**, msad211 (2023).
41. J. R. Adion, J. G. Galloway, A. D. Kern, Predicting the landscape of recombination using deep learning. *Mol. Biol. Evol.* **37**, 1790–1808 (2020).
42. C. Battey, G. C. Coffing, A. D. Kern, Visualizing population structure with variational autoencoders. *G3 (Bethesda)* **11**, jkaa036 (2021).
43. Z. Mo, A. Siepel, Domain-adaptive neural networks improve supervised machine learning based on simulated population genetic data. *PLoS Genet.* **19**, e1011032 (2023).
44. A. Radford, K. Narasimhan, Improving language understanding by generative pre-training. (2018).
45. G. Benegas, C. Albers, A. J. Aw, C. Ye, Y. S. Song, A DNA language model based on multispecies alignment predicts the effects of genome-wide variants. *Nat. Biotechnol.* **43**, 1960–1965 (2025).
46. G. Brix, et al., Genome modeling and design across all domains of life with Evo 2. *bioRxiv* pp. 2025–02 (2025).
47. 1000 Genomes Project Consortium, A map of human genome variation from population scale sequencing. *Nature* **467**, 1061 (2010).
48. A. Auton, T. Salcedo, The 1000 genomes project in *Assessing Rare Variation in Complex Traits*. (Springer), pp. 71–85 (2015).
49. Anopheles gambiae 1000 Genomes Consortium, Genetic diversity of the African malaria vector Anopheles gambiae. *Nature* **552**, 96 (2017).
50. C. S. Clarkson, et al., Genome variation and population structure among 1142 mosquitoes of the African malaria vector species Anopheles gambiae and Anopheles coluzzii. *Genome Res.* **30**, 1533–1546 (2020).
51. Y. Deng, Y. S. Song, R. Nielsen, A general framework for branch length estimation in ancestral recombination graphs. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2504461122 (2025).
52. X. Grau-Bové, et al., Evolution of the insecticide target Rdl in African Anopheles is driven by interspecific and interkaryotypic introgression. *Mol. Biol. Evol.* **37**, 2900–2917 (2020).
53. D. M. Swallow, Genetics of lactase persistence and lactose intolerance. *Annu. Rev. Genet.* **37**, 197–219 (2003).
54. S. A. Tishkoff, et al., Convergent adaptation of human lactase persistence in Africa and Europe. *Nat. Genet.* **39**, 31–40 (2007).
55. T. Bersaglieri, et al., Genetic signatures of strong recent positive selection at the lactase gene. *Am. J. Hum. Genet.* **74**, 1111–1120 (2004).
56. Y. Itan, A. Powell, M. A. Beaumont, J. Burger, M. G. Thomas, The origins of lactase persistence in Europe. *PLoS Comput. Biol.* **5**, e1000491 (2009).
57. B. M. Peter, E. Huerta-Sanchez, R. Nielsen, Distinguishing between selective sweeps from standing variation and from a de novo mutation. *PLoS Genet.* **8**, e1003011 (2012).
58. P. W. Hedrick, Balancing selection and MHC. *Genetica* **104**, 207–214 (1998).
59. L. Azevedo, C. Serrano, A. Amorim, D. N. Cooper, Trans-species polymorphism in humans and the great apes is generally maintained by balancing selection that modulates the host immune response. *Hum. Genomics* **9**, 21 (2015).
60. A. L. Fortier, J. K. Pritchard, Ancient trans-species polymorphism at the major histocompatibility complex in primates. *eLife* **14**, RP103547 (2025).
61. R. R. Love, et al., Chromosomal inversions and ecotypic differentiation in Anopheles gambiae: the perspective from whole-genome sequencing. *Mol. Ecol.* **25**, 5889–5906 (2016).
62. M. C. Fontaine, et al., Extensive introgression in a malaria vector species complex revealed by phylogenomics. *Science* **347**, 1258524 (2015).
63. R. H. French-Constant, T. A. Rocheleau, J. C. Steichen, A. E. Chalmers, A point mutation in a Drosophila GABA receptor confers insecticide resistance. *Nature* **363**, 449–451 (1993).
64. M. Thompson, J. Steichen, R. French-Constant, Conservation of cyclodiene insecticide resistance-associated mutations in insects. *Insect Mol. Biol.* **2**, 149–154 (1993).
65. W. Du, et al., Independent mutations in the Rdl locus confer dieldrin resistance to Anopheles gambiae and An. arabiensis. *Insect Mol. Biol.* **14**, 179–183 (2005).
66. R. Elliott, V. Ramakrishna, Insecticide resistance in Anopheles gambiae Giles. *Nature* **177**, 532–533 (1956).
67. G. Davidson, J. Hamon, A case of dominant dieldrin resistance in Anopheles gambiae Giles. *Nature* **196**, 1012–1012 (1962).
68. F. Baumdicker, et al., Efficient ancestry and mutation simulation with msprime 1.0. *Genetics* **220**, iyab229 (2022).
69. A. Radford, et al., Language models are unsupervised multitask learners. (2019).
70. A. Vaswani, et al., Attention is all you need (2017).
71. J. Su, et al., RoFormer: Enhanced transformer with rotary position embedding (2023) arXiv:2104.09864.